USN | | | | | | | | | | [24MTAI12]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

KNOWLEDGE REPRESENTATION AND REASONING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.4
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Define the Types of knowledge and discuss examples of tasks that require both declarative and procedural knowledge?	**1**	**2**	**10**
	b.	Elucidate techniques for representing knowledge in expert systems such as rules, frames, or ontologies?	**1**	**2**	**10**
2.	**a.**	Define and discuss types of modal logic (e.g., S4, S5), and how do they differ in their assumptions about necessity and possibility?	**2**	**3**	**10**
	b.	Compare and contrast Deductive reasoning, Inductive reasoning and Abductive reasoning with an example.	**2**	**4**	**10**
3.	**a.**	Illustrate the components of ontologies in detail	**3**	**3**	**10**
	b.	How does Protégé, as a popular ontology editor, support ontology construction through features like class hierarchy, relationships and reasoning?	**3**	**3**	**10**
		OR			
4.	**a.**	Analyse the role of Semantic Web. How does RDF represent data using triples (subject-predicate-object)?	**3**	**4**	**10**
	b.	Infer the challenges encountered in the ontology alignment process in a specific case. Explain with real-world cases in terms of best practices for successful ontology integration.	**3**	**4**	**10**
5.	**a.**	Analyse Markov model. In respect of Markov chains, including states, transitions, and the Markov property. Also explain advantages and limitations of using Markov models for modeling sequences or time series data.	**4**	**4**	**10**
	b.	Analyse fuzzy inference, from fuzzy inputs to generate fuzzy outputs. Explain the main steps involved in fuzzy inference, including fuzzification, rule evaluation, and defuzzification?	**4**	**4**	**10**
		OR			
6.	**a.**	Examine the concept of Non-Monotonic Reasoning with a case study in detail.	**4**	**4**	**10**
	b.	Compare Classical vs. probabilistic vs. non-monotonic reasoning with an example.	**4**	**3**	**10**

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7.	**a.**	Examine the Google Knowledge Graph to enhance the search experience for users, with the different set of information.	**5**	**5**	**10**
	b.	Construct ACT-R, with an aim to simulate human cognition through its modular architecture.	**5**	**3**	**10**
		OR			
8.	**a.**	Solve the model-agnostic methods for explaining predictions, such as LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations).	**6**	**3**	**10**
	b.	Illustrate current trends in the development of knowledge representation languages, and how are they evolving to meet the needs of modern applications?	**6**	**3**	**10**